

Solutions: Additional Exercises 1

Limits & Continuity of Multivariable Functions

Exercise 1: Domains

- (a) $f(x, y) = \ln(x^2 - y)$. Need $x^2 - y > 0$, i.e. $y < x^2$. **Domain:** the open region below the parabola $y = x^2$.
- (b) $f(x, y) = \sqrt{9 - x^2 - y^2}$. Need $9 - x^2 - y^2 \geq 0$, i.e. $x^2 + y^2 \leq 9$. **Domain:** the closed disk of radius 3 centered at the origin.
- (c) $f(x, y) = \arcsin\left(\frac{x+y}{x-y}\right)$. Need $-1 \leq \frac{x+y}{x-y} \leq 1$ and $x \neq y$. From $\frac{x+y}{x-y} \leq 1$: $x+y \leq x-y$ (when $x > y$), giving $y \leq 0$. From $\frac{x+y}{x-y} \geq -1$: $x+y \geq -(x-y) = y-x$, giving $x \geq 0$. **Domain:** $\{(x, y) : x \geq 0, y \leq 0, x \neq y\}$ (third quadrant boundary plus positive x -axis, excluding $x = y$, which is only $x = y = 0$ in this region so we must also check). More carefully: the region is $x \geq 0, y \leq 0$ with $x \neq y$. Since $y \leq 0 \leq x$ we have $x \geq y$ so $x - y \geq 0$, and $x = y$ only at origin which is a boundary case. Domain: $x \geq 0, y \leq 0$.
- (d) $f(x, y) = \ln\left(\frac{x^2+y^2}{x^2-y^2}\right)$. Need $\frac{x^2+y^2}{x^2-y^2} > 0$. Since $x^2 + y^2 \geq 0$ (and $= 0$ only at origin), we need $x^2 - y^2 > 0$, i.e. $|x| > |y|$. Also need $x^2 + y^2 \neq 0$, so not the origin. **Domain:** $\{(x, y) \neq (0, 0) : |x| > |y|\}$ (the two open sectors between the lines $y = \pm x$).
- (e) $f(x, y) = \sqrt{\ln(x^2 + y^2 - 1)}$. Need $\ln(x^2 + y^2 - 1) \geq 0$, i.e. $x^2 + y^2 - 1 \geq 1$, so $x^2 + y^2 \geq 2$. Also need $x^2 + y^2 - 1 > 0$, which is implied. **Domain:** $\{(x, y) : x^2 + y^2 \geq 2\}$ (exterior of circle of radius $\sqrt{2}$, including the circle).

Exercise 2: Level Curves

- (a) $f(x, y) = x^2 - y^2 = c$. Level curves are hyperbolas $x^2 - y^2 = c$. For $c > 0$: hyperbola opening left-right. For $c < 0$: hyperbola opening up-down. For $c = 0$: the lines $y = \pm x$.
- (b) $f(x, y) = y/x = c$. Level curves are lines $y = cx$ through the origin (excluding origin). Each slope c gives one line; $c = 0$ is the x -axis (excluding origin).
- (c) $f(x, y) = x^2 + 4y^2 = c$ (need $c > 0$). These are ellipses with semi-axes $\sqrt{c}$ along x and $\sqrt{c}/2$ along y . At $c = 4$: semi-axes 2 and 1. At $c = 16$: semi-axes 4 and 2.
- (d) $f(x, y) = \ln(x^2 + y^2) = c$ means $x^2 + y^2 = e^c > 0$. Level curves are concentric circles of radius $e^{c/2}$. At $c = 0$: unit circle. At $c = 2$: circle of radius e . At $c = -2$: circle of radius e^{-1} .

Exercise 3: Limits

- (a) Let $t = x^2 + y^2 \rightarrow 0^+$. Then $\lim_{t \rightarrow 0^+} \frac{\sin t}{t} = 1$. **Answer:** 1. (Standard limit.)
- (b) Use polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, $r \rightarrow 0^+$: $\frac{2r^2 \cos^2 \theta + 3r^4 \sin^4 \theta}{r^2} = 2 \cos^2 \theta + 3r^2 \sin^4 \theta \rightarrow 2 \cos^2 \theta$. This depends on θ , so the **limit does not exist**.
- (c) Since $|\cos(1/xy)| \leq 1$, we have $|(x^2 + y^2) \cos(1/xy)| \leq x^2 + y^2 \rightarrow 0$. By the squeeze theorem, **limit** = 0.
- (d) Try paths: Along $x = 0$: $f(0, y) = 0$. Along $y = x^2$: $f(x, x^2) = \frac{x^3 \cdot x^2}{x^4 + x^4} = \frac{x^5}{2x^4} = \frac{x}{2} \rightarrow 0$. Hmm, both give 0. Try $y = tx^2$ general: $f(x, tx^2) = \frac{x^3 \cdot tx^2}{x^4 + t^2 x^4} = \frac{tx}{1+t^2} \rightarrow 0$. All paths give 0. Actually let's check more carefully via polar: $\frac{r^3 \cos^3 \theta \cdot r \sin \theta}{r^4 \cos^4 \theta + r^2 \sin^2 \theta} = \frac{r^4 \cos^3 \theta \sin \theta}{r^4 \cos^4 \theta + r^2 \sin^2 \theta}$. If $\sin \theta \neq 0$, denominator $\sim r^2 \sin^2 \theta$ as $r \rightarrow 0$, so ratio $\sim \frac{r^4 \cos^3 \theta \sin \theta}{r^2 \sin^2 \theta} = \frac{r^2 \cos^3 \theta}{\sin \theta} \rightarrow 0$. If $\sin \theta = 0$ (axes), numerator = 0.

Actually, there IS a path where the limit is nonzero. Let $y = x^2$: computed above gives $x/2 \rightarrow 0$. Let me try $y = x^{4/3} \cdot m \dots$ Actually we need to check the original hint that DNE via path $y = x^2$. But I computed $f(x, x^2) = \frac{x^5}{2x^4} = x/2 \rightarrow 0$. So via that path limit is 0.

The original note says this is DNE (path $y = x^2$ gives 1/2), but that was for a different function. Let's recompute: $\frac{x^3 y}{x^4 + y^2}$ at $y = x^2$: $\frac{x^3 \cdot x^2}{x^4 + x^4} = \frac{x^5}{2x^4} = \frac{x}{2} \rightarrow 0$. Not 1/2.

Path $y = mx^2$: $\frac{x^3 \cdot mx^2}{x^4 + m^2 x^4} = \frac{mx^5}{x^4(1+m^2)} = \frac{mx}{1+m^2} \rightarrow 0$. All such paths give 0.

Polar: As computed, limit is 0 as well. **Answer:** **limit** = 0.

(This problem's limit exists and equals 0; the "DNE" comment in the problem statement referred to a similar but different function.)

- (e) Rationalize: multiply by $\frac{\sqrt{x^2+y^2+1}+1}{\sqrt{x^2+y^2+1}+1}$:

$$\frac{(x^2 + y^2)(\sqrt{x^2 + y^2 + 1} + 1)}{x^2 + y^2} = \sqrt{x^2 + y^2 + 1} + 1 \rightarrow \sqrt{1} + 1 = 2.$$

Answer: 2.

- (f) Write $t = x^2 + y^4$. If $t \rightarrow 0$ implies $(x, y) \rightarrow (0, 0)$: $\frac{\sin t}{x^2 + y^2}$. Note $x^2 + y^2 \geq 0$ and $x^2 \leq x^2 + y^4$ but we need to relate $t = x^2 + y^4$ to $x^2 + y^2$. Near $(0, 0)$: $y^4 \leq y^2$ so $x^2 + y^4 \leq x^2 + y^2$. Also $x^2 + y^2 \leq \text{something}$. But $\frac{\sin(x^2+y^4)}{x^2+y^2} \approx \frac{x^2+y^4}{x^2+y^2}$ as $(x, y) \rightarrow (0, 0)$. Along $y = 0$: $\frac{x^2}{x^2} = 1$. Along $x = 0$: $\frac{y^4}{y^2} = y^2 \rightarrow 0$. Two paths give different limits, so **limit does not exist**.
- (g) $(1 + \frac{1}{x})^{x+y} = (1 + \frac{1}{x})^x \cdot (1 + \frac{1}{x})^y$. As $x \rightarrow \infty$: $(1 + \frac{1}{x})^x \rightarrow e$ and $(1 + \frac{1}{x})^y \rightarrow 1^3 = 1$ (for fixed $y \rightarrow 3$). **Answer:** e .
- (h) As $(x, y) \rightarrow (0, 0)$ with $xy \neq 0$: $\frac{\sin(xy)}{xy} \rightarrow 1$, but $\sin\left(\frac{1}{\sqrt{x^2+y^2}}\right)$ oscillates. Take $x = t$, $y = t$ with $t \rightarrow 0^+$: $\frac{\sin(t^2)}{t^2} \cdot \sin\left(\frac{1}{t\sqrt{2}}\right)$. The first factor $\rightarrow 1$, while the second oscillates between -1 and 1 . **The limit does not exist**.

- (i) Polar coordinates: $\frac{r^5 \cos^5 \theta - r^5 \cos^2 \theta \sin^3 \theta}{r^4 \cos^4 \theta + r^4 \sin^4 \theta} = \frac{r \cos^2 \theta (\cos^3 \theta - \sin^3 \theta)}{\cos^4 \theta + \sin^4 \theta} \rightarrow 0$. (The denominator $\cos^4 \theta + \sin^4 \theta \geq 1/2$, and the numerator has factor $r \rightarrow 0$.) **Answer:** 0.
- (j) $\cos(x^2 + y^2) \approx 1 - \frac{(x^2 + y^2)^2}{2}$ near origin, so $1 - \cos(x^2 + y^2) \approx \frac{(x^2 + y^2)^2}{2}$. Thus $\frac{x^5 - \cos(x^2 + y^2) + 1}{(x^2 + y^2)^2} \approx \frac{x^5}{(x^2 + y^2)^2} + \frac{1}{2}$. The term $\frac{x^5}{(x^2 + y^2)^2}$: in polar, $\frac{r^5 \cos^5 \theta}{r^4} = r \cos^5 \theta \rightarrow 0$. **Answer:** $\frac{1}{2}$.

Exercise 4: Continuity

- (a) Polar: $\frac{r^2 \cos^2 \theta + 2r^2 \sin^2 \theta}{r^2} = \cos^2 \theta + 2\sin^2 \theta = 1 + \sin^2 \theta$. This depends on θ , so $\lim_{(x,y) \rightarrow (0,0)} f \neq 1 = f(0,0)$. **Not continuous at (0,0). Domain of continuity:** $\mathbb{R}^2 \setminus \{(0,0)\}$.
- (b) Polar: $\frac{r^4 \cos^2 \theta \sin^2 \theta}{r^2} = r^2 \cos^2 \theta \sin^2 \theta \rightarrow 0 = f(0,0)$. **Continuous on all $\mathbb{R}^2$.**
- (c) $\left| \frac{\sqrt[3]{y} \sin(x^2)}{x^2 + y^2} \right| \leq \frac{|y|^{1/3} \cdot x^2}{x^2 + y^2} \leq |y|^{1/3} \rightarrow 0 = f(0,0)$ (used $x^2/(x^2 + y^2) \leq 1$). **Continuous on all $\mathbb{R}^2$.**
- (d) $|(x + y) \sin(1/x^2) \cos(1/y)| \leq |x + y| \leq |x| + |y| \rightarrow 0 = f(0,0)$. **Continuous on all $\mathbb{R}^2$.**
- (e) Polar: $\frac{r^3 \cos^3 \theta + r^3 \sin^3 \theta}{r^2} = r(\cos^3 \theta + \sin^3 \theta) \rightarrow 0 = f(0,0)$. (Bounded: $|\cos^3 \theta + \sin^3 \theta| \leq 2$.) **Continuous on all $\mathbb{R}^2$.**
- (f) Polar numerator: $r^4 \cos^4 \theta \cdot r^2 \sin^2 \theta = r^6 \cos^4 \theta \sin^2 \theta$. Denominator: $r^4 \cos^4 \theta + r^2 \sin^2 \theta = r^2(r^2 \cos^4 \theta + \sin^2 \theta)$. Ratio: $\frac{r^4 \cos^4 \theta \sin^2 \theta}{r^2 \cos^4 \theta + \sin^2 \theta}$. If $\sin \theta \neq 0$: denom $\rightarrow \sin^2 \theta$, so ratio $\rightarrow \frac{r^4 \cos^4 \theta \sin^2 \theta}{\sin^2 \theta} \rightarrow 0$. If $\sin \theta = 0$: ratio = 0. So limit = 0 = $f(0,0)$. **Continuous on all $\mathbb{R}^2$.**
- (g) Use polar: numerator $x^4 - x^3 y - x y^3 + y^4 = (x - y)(x^3 - y^3) = (x - y)^2(x^2 + xy + y^2)$ (after factoring). In polar: $(r \cos \theta - r \sin \theta)^2(r^2 \cos^2 \theta + r^2 \cos \theta \sin \theta + r^2 \sin^2 \theta) = r^2(\cos \theta - \sin \theta)^2 \cdot r^2(1 + \cos \theta \sin \theta)$. Denominator: $\sqrt{r^6 \cos^6 \theta + r^6 \sin^6 \theta} = r^3 \sqrt{\cos^6 \theta + \sin^6 \theta}$. Ratio: $\frac{r^4(\cos \theta - \sin \theta)^2(1 + \cos \theta \sin \theta)}{r^3 \sqrt{\cos^6 \theta + \sin^6 \theta}} = r \cdot \frac{(\cos \theta - \sin \theta)^2(1 + \cos \theta \sin \theta)}{\sqrt{\cos^6 \theta + \sin^6 \theta}}$. The fraction is bounded (denom $\geq 1/\sqrt{2}$, numerator bounded). So ratio $\rightarrow 0 = f(0,0)$. **Continuous on all $\mathbb{R}^2$.**
- (h) $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \sin \frac{1}{x^2 + y^2}$: polar: $r^2 \sin(1/r^2) \rightarrow 0$ (since $|\sin| \leq 1$). But $f(0,0) = 1 \neq 0$. **Not continuous at (0,0). Domain of continuity:** $\mathbb{R}^2 \setminus \{(0,0)\}$.